Empathetic AI

UCI MDS Capstone Final Report in Partnership with Humma.AI

Author List:
Raunak Srivastava
Akhil Krosuru Mohan
Avi Tewari
Vignaya Virajith Anumula
Megha Sri Satya Sai Devineni
Namrata Nyamagoudar

Collaborator Information
Humma.AI - Katie Smith (CEO and Co-Founder of Humma.AI)

Executive Summary

- Humma.AI is a startup creating a platform to connect users through shared experiences and stories while integrating an Empathetic AI to foster ethical online interactions.
- The Humma.AI Capstone Team's goal is to develop an Empathetic AI model that provides culturally competent responses.
- The team fine-tuned Meta's Llama 3.2 3B and Mistral 7B models using open-source Mental Health Conversations, Empathetic Dialogues, and GoEmotions datasets.
- Improved empathy metrics, focusing on EQ-60 scores, through parameter-efficient fine-tuning techniques such as LoRA.
- Built a Streamlit web application with chat history enabled to allow follow-up messages.
- Showed significant improvements in empathy levels using EQ-60, SQ-R, and AEQr benchmarks.
- Human evaluators are used to test responses from our Empathetic AI vs. other chatbots.
- The next steps include aligning the UI with Humma.AI's Figma design, contacting Humma.AI's focus group volunteers, and shifting the model focus to cultural competence.

Overview & Project Goals

Humma.AI is a Los Angeles-based startup with the goal of creating a storytelling and discussion platform to connect individuals with similar stories. It aims to provide an outlet for people to express themselves through micro-posts, audio, video, and images. Its core mission is to create an AI that not only understands but also responds to human emotions in real-time, leading to more meaningful and engaging interactions. Humma.AI stands at the forefront of the rapidly evolving AI landscape by focusing on integrating emotional intelligence into traditional AI frameworks.

Our project goal is to develop an Empathetic AI model that can provide ethical, culturally competent responses to user stories. We plan to host this model on AWS and seamlessly integrate it with a chatbot UI for Humma.AI's users. By providing access to a platform where users can share their experiences, the AI model would act as an empathetic companion, bridging the gap between technology and human connection.

Background Information

Before diving further into our own goals, it is important to fully understand the problem Humma.AI has identified; this revolves around the current state of social platforms, particularly their challenges concerning user friendliness and satisfaction. A similar theme amongst existing platforms has been their prioritization of algorithms that yield maximum user engagement. The downside to such an approach is the amplification of biases and misinformation, many times fostered in toxic communities where hate speech and divisive content thrive.

After thinking more profoundly about this cause-and-effect, it makes sense why spaces with more visibility are those centered around controversy. With the increasing polarization of social platform environments, users have responded in various ways. While some users have maintained a high presence in these environments, others have been left to voice their feelings, ideas, & opinions. In the case of Humma.AI, they aim to protect user "stories". The fear of

backlash has left these spaces void of the authentic user engagement that held focus during the inception of left social platforms. The feeling of being disconnected from supportive communities has caused a decline in user satisfaction of many social platforms. With this in mind, Humma.AI has vaulted itself as a "storytelling and discussion network for creating, sharing, and discovering people with similar stories to get connected, validated, and a sense of belonging."

Another consequence of growing biases has been user distrust, specific to platforms that regularly collect users' personal data without two-sided transparency. In response, Humma.AI promises "to fix the bias and discrimination in AI and make sure the data is controlled by us, not big tech." Assessing the relevance of Humma.AI's mission among tech experts, DataRobot's State of AI Bias Report revealed that "many leaders share deep concerns around the risk of bias in AI (54%) and a growing desire for government regulation to prevent bias in AI (81%)" based on an exploration of over 350 organizations across industries (DataRobot, 2022).

In addition to fixing the bias and discrimination in AI, Humma.AI is "building an Empathetic AI system to guide each of us in creating our stories, linking us to similar stories, plus more." From this, we have captured our previously stated project goal & contribution to Humma.AI: developing an Empathetic AI model that can give ethical, culturally competent responses to user stores.

Key Terms/Concepts:

- EmpatheticAI: A modern approach to artificial intelligence that emphasizes understanding human emotions and experiences, thereby creating a safer and more inclusive digital environment.

- Cultural Competence: The ability to understand and effectively communicate with people from diverse beliefs and backgrounds. Vital steps towards cultural competence include developing awareness of one's own biases, and in response, adapting behaviors to minimize the visibility of those biases.

- Bias in AI: Systematic unfairness in AI systems due to a variety of factors, some being prejudiced data, flawed algorithms, and imbalanced training.

Relevant EDA

With one of our main goals being the creation of an Empathetic AI model that can give ethical, culturally competent responses to user stores, it is important to see how current AI has performed in similar tasks. Existing psychological test measures are used as a benchmark to evaluate GenAI models on their empathetic capability.

The following tests were used:

- The Empathy Quotient **(EQ-60)** measures the ability to feel (affective) an appropriate emotion in response to another's emotion and the ability to understand (cognitive) another's emotion. It's a self-report test consisting of 60 statements answered using the four options: "strongly agree", "slightly agree", "slightly disagree", or "strongly disagree". The highest achievable score is 80.
- The Systemizing Quotient Revised **(SQ-R)** tests the desire for and the ability to conduct systemized thinking. (In the context of the Empathizing-Systemizing (E-S) theory, people with lower SQ-R scores may lean more toward empathizing, i.e., focusing on understanding and responding to other people's feelings.) It is also a self-reporting test

consisting of 75 user statements answered using the five options: “strongly agree”, “slightly agree”, “slightly disagree”, “strongly disagree”, and “neither agree nor disagree”. The highest achievable score is 80.

- The Applied Empathy Quotient Ratio **(AEQr)** is a hypothetical metric designed to measure the capacity to apply empathy in situations where humans, despite having a rational understanding of a feeling, face challenges in genuinely empathizing. It is calculated as:

$$AEQr = (EQ - 60)/(SQ - R)$$

Having an AEQr ratio of >= 1.3 tends to acknowledge someone as appearing more empathetic as a whole

Model	Type	EQ-60	SQ-R	AEQr
*Human Adult Female		47	24	1.95
*Human Adult Male		42	30	1.40
ChatGPT 4 turbo-preview	open	36	34	1.05
Claude v2	public	44	28	1.60
Claude v3 Opus	public	56	75	0.75
Gemini 1.0	public	55	53	1.04
Llama 70B	open	67	61	1.10
Mistral Large	public	38	54	0.70
Mixtral-8x7b-32768	open	39	54	0.72
Pi.ai (not via API)	closed	64	50	1.20
Willow v1	closed	61	31	1.97

Fig 1:Psychological test measures for pubic (open and closed) models

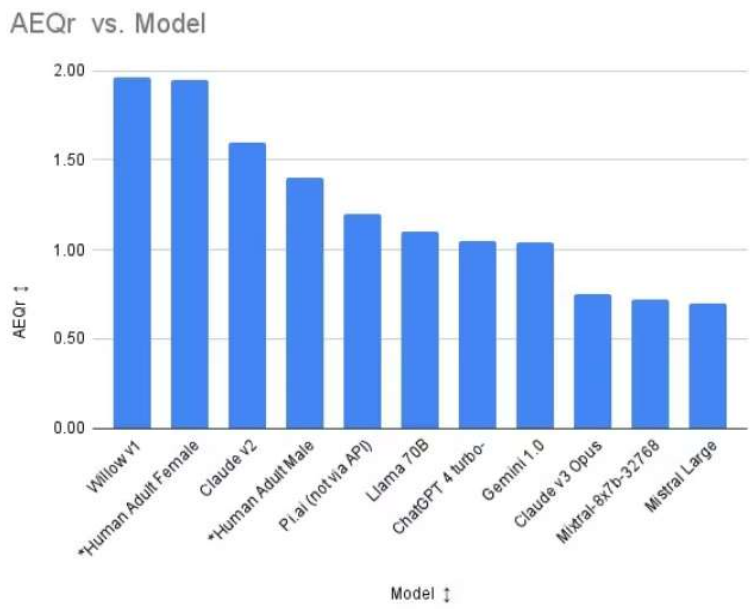


Fig 2: AEQr scores of the models

From Figure 1, we observe that raw LLMs fail to connect with users empathetically (even though they use empathetic phrasing, EQ-60 is balanced out by their high SQ-R scores). Popular models like Claude v3, Llama 70B, and Pi.ai have record-high EQ-60 scores, but this is offset by high SQ-R scores. The takeaway from this is that these models show questionable performance in understanding emotions and applying them to dialog generation. Interestingly, Willow performed exceptionally well, as indicated by its high AEQr ratio—the highest among its peers—despite developers observing that the model prioritizes emotions over logic and facts when deemed necessary.

System Design

Our approach involved a comprehensive system design process. We aggregated datasets from various sources, storing them in AWS S3 buckets for easy access among team members. These datasets were then consolidated and divided into training and testing sets. Considering the computationally expensive nature of training large language models (LLMs) from scratch, we opted to fine-tune existing LLMs. We explored several models, including Meta's LLama series, Microsoft's Phi series, and Mistral, while simultaneously experimenting with user interface-driven fine-tuning solutions from AWS Sagemaker Jumpstart and AWS Bedrock.

Through Humma.AI's partnership with AWS, we received access to a range of computing resources. Initially, we utilized the p3.8xlarge instance with four Tesla V100 GPUs, each equipped with 16GB of memory. This setup, however, proved insufficient for even smaller models, prompting a shift to the larger G5.24xlarge instance with four Nvidia A10g GPUs, each boasting 24GB of memory. Despite increasing compute capacity, memory constraints persisted, necessitating the adoption of parameter-efficient fine-tuning techniques such as LoRA and fast training frameworks like Unsloth, Axolotl, and DeepSpeed. Leveraging multi-GPU support and offloading certain computations to CPUs, we successfully fine-tuned smaller models like the LLama 3.2 1B parameters and Phi Mini. Prompt engineering was a pivotal component in this project, enabling us to shape the LLM's conversational abilities effectively. By defining clear roles, characteristics, and guidelines within the system prompt, we were able to enhance the model's capacity to provide empathetic and engaging responses. This approach proved to be an efficient strategy to meet the project's objectives of reducing implicit bias and increasing levels of engagement in AI interactions.

We noticed a considerable rise in the EQ-60 scores when we increased the neural network complexity from 1B to 3B parameters, prompting us to explore fine-tuning for larger models like Mistral 7B and Llama 3.1 8B. To accommodate these larger models, we upgraded to the P4d.24xlarge instance featuring eight Nvidia A100 GPUs, each with 40GB of memory. Unsloth is a fast training framework that tweaks the neurons at the matrix level by introducing manual quantizations to the weights of internal neurons and connections. Utilizing Unsloth's capability for faster training times and reduced memory usage, we fine-tuned these models efficiently. Although validation scores across architectures were similar, given the textual nature of our data, we could not solely rely on evaluation losses to measure the emotional responsiveness of our models. Thus, we implemented real-world empathy evaluators like EQ-60, SQ-R, AEQr, human evaluators, and weighted empathy scores to effectively gauge our models' empathetic capabilities. These models were later moved into S3 buckets to create endpoints for us to access from the web UI. For the time being, we've built a simple Streamlit web application that takes the user story as input and returns an empathetic response. The UI enables chat history so the user can enjoy an interactive experience, allowing our model to respond to follow-up messages while remembering the context and historical interaction.

Tradeoffs or Alternative Approaches

For the training data, our project partners' vision was to use user stories to train the model. But we were stuck in the chicken and egg problem – we could not collect user stories unless we had a base model for them to interact with, but we could not develop this base model without user stories. To overcome this, we decided to collect our own data to train the model. We collected data from 3 major datasets: Empathetic Dialogues (published by Meta Research), GoEmotions (published by Google Research), and Mental Health Conversations, a context-response dataset

covering a wide range of mental health topics whose responses are provided by qualified psychologists.

We also explored the AWS Sagemaker Jumpstart Application, which allows users to perform fine-tuning with their custom data in either a low-code or no-code (UI-based) environment. However, the biggest tradeoff with this approach was that we needed to use a very specific data format which required a lot of trial and error to achieve and still provided us with dissatisfying performance. Additionally, the approach is relatively new, and the limited documentation did not provide much assistance when we ran into issues. Another challenge we faced was the need for access to resources from the AWS Quota team, which forced us to delay our deadlines until the resource requests were approved.

Using our existing computational resources, we also explored another LLM framework called Llama-Factory. This framework also offers a similar UI application hosted on Gradio, which allows users to pull models from Hugging Face and fine-tune them with datasets provided by the Llama-Factory team. However, the major issue we encountered with running Llama-Factory on AWS was the need to set up a custom Virtual Private Cloud (VPC) to host the Gradio application. This requirement extended somewhat beyond the scope of our academic project, and even if addressed externally, we were still constrained in the types of fine-tuning tasks we could perform.

Validation

For validation, we took two approaches – quantifying empathy with a weighted "Empathy Score" and using human evaluators. We largely focused on human evaluation, as it aligns with Humma.AI's product vision for incorporating user feedback.

To calculate Empathy Score, we took a weighted average of Sentiment Analysis using VADER, Emotion Analysis using the DistilRoBERTa model to assign probability scores, and Engagement score, a metric used to evaluate a response's engagement with a pre-defined dictionary of empathetic words. The final Empathy Score was calculated by weighting 40% Sentiment, 40% Emotion, and 20% Engagement. Below is a figure of Empathy Scores calculated on our model versus ChatGPT on the same prompts.

*Empathy Score=0.4*Sentiment Score+0.4*Empathy Emotion Score+0.2*Phrase Score*

Sentiment Score:

$$\text{Sentiment Score} = \text{Positive Sentiment} - \text{Negative Sentiment}$$

Empathy Emotion Score:

$$\text{Empathy Emotion Score} = \sum_{i=1}^{n} \text{Emotion Score}_i \quad \text{where} \quad \text{Emotion Label}_i \in \{\text{"compassion"}, \text{"caring"}, \text{"empathy"}\}$$

Phrase Score:

$$\text{Phrase Score} = \sum_{j=1}^{m} I(\text{Empathy Phrase}_j \in \text{Response})$$

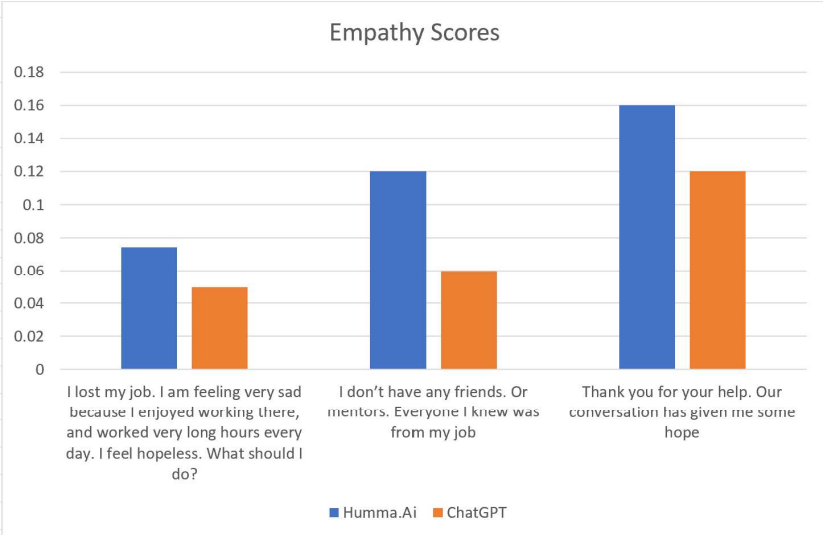


Fig 3: Empathy Scores generated for the responses in Google Form

Model	EQ-60	SQ-R	AEQr
LLama 3.2 3B (base)	49	47	~1.05
LLama 3.2 3B (FT)	53	41	~1.3

Fig 4: Psychological test scores between base and fine-tuned Llama 3.2 3B

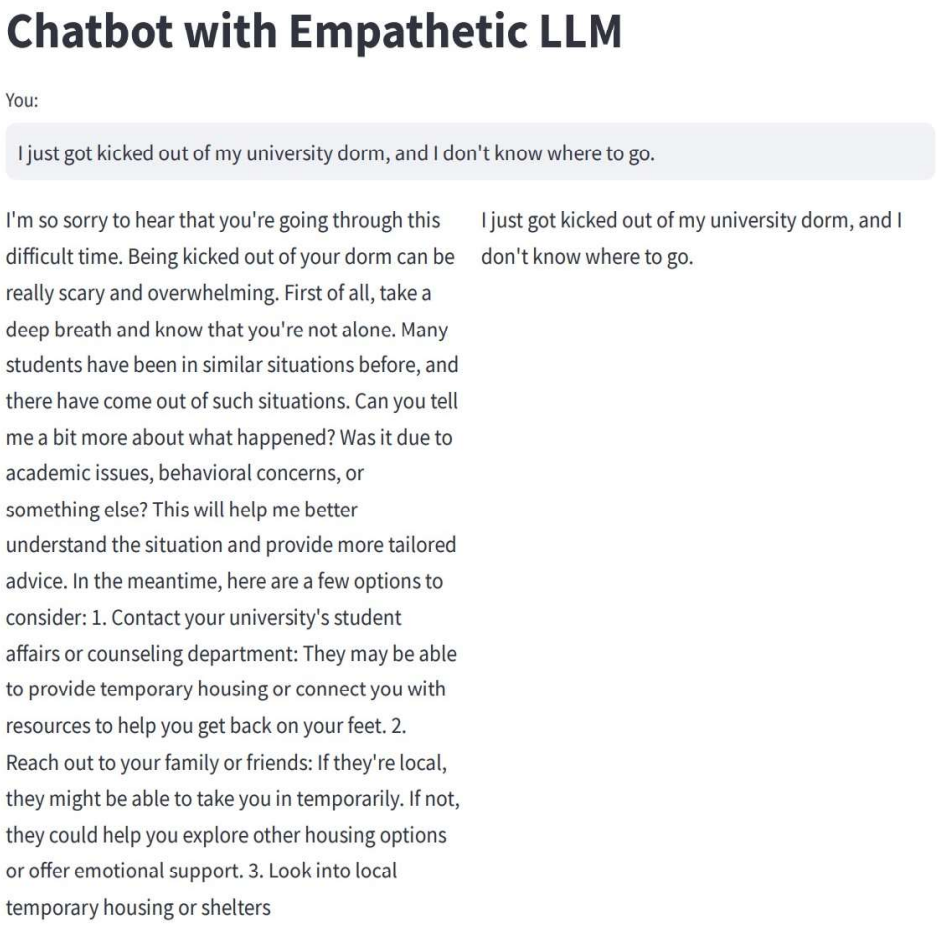
Returning to the psychological test scores, we evaluated whether our approach enhanced the model's understanding of empathy by conducting the aforementioned tests on both the base and fine-tuned versions of Llama 3.2 3B. We picked and presented Llama 3.2 because we found that this model has produced significant results while keeping the computation time low compared to the other models with higher parameter counts. For reference, the Mistral 7B scored 48/80 and Llama 3.1 8B scored 51/80 in the EQ-60 category. (the SQ-R scores for both were very similar to the base, but the model was inconsistent most of the time, due to the nature of the user statements – instead of confining responses to the given options, the models often produced vague responses like, *"I'm not sure how to answer this"* or *"There are multiple ways to look at it"* and so on, even after prompting. As a result, we chose not to record these scores.)

To further validate the empathetic capabilities of the fine-tuned Llama 3.2 3B model, we conducted human evaluation using a Google Form survey to collect feedback. After running demo inputs shown in Figure 3, we compiled the responses given by our model and those from ChatGPT into a document which was sent to 22 participants. While we didn't direct attention to the makeup of our demo group, our main focus was on whether humans would prefer our model's output as being the more empathetic response. When asked "Was our chatbot more

empathetic than other chatbots like ChatGPT?", 81.8% of participants replied 'Yes' with 'No' and 'Unsure' as other options. When asked "Which chatbot provided responses that felt closer to what you'd expect from a human?" and forced to choose between the two options, 90.9% of participants chose our model. When asked to rate the depth of responses, 54.5% of participants said our chatbot was much deeper, 27.3% said our chatbot was slightly deeper, and the remaining 18.2% were spread amongst the remaining 3 options (all strayed further away from preferring our model). When presented with a 1 through 5 inclusive integer scale, our participants gave us an average response of 4.18 stars when asked to rate the helpfulness of our chatbot and 4.27 stars when asked to rate its overall empathy.

Demo

Below is a comparison of our UI for Empathetic AI and ChatGPT. Our responses are more empathetic than the generic responses provided by ChatGPT. Our model asks follow-up questions to deliver responses that are more considerate and empathetic.



Vs

I just got kicked out of my university dorm, and I don't know where to go.

I'm so sorry you're dealing with this. Here are some steps to help you find a solution quickly:

1. **Reach Out to University Services**: Contact your university's housing office, student affairs, or emergency assistance services. They might have temporary housing options or resources for displaced students.
2. **Friends and Community**: Reach out to friends, classmates, or professors who may be able to offer temporary accommodations or connect you with someone who can.
3. **Local Shelters or Short-Term Rentals**: Search for shelters, hostels, or short-term rental platforms like Airbnb. Some shelters offer special assistance for students.
4. **Financial Assistance**: If funds are an issue, look into emergency student funds, loans, or grants offered by your university or local organizations.
5. **Campus Organizations**: Check with campus organizations, clubs, or religious groups. They often have networks or resources to help in situations like this.

Fig 5: Responses in our prototype & ChatGPT

Chatbot with Empathetic LLM

You:

I am in my last semester of grad school, and I failed a course. I feel very depressed because this failure m

I'm so sorry to hear that you're going through this challenging time. It's completely understandable to feel overwhelmed and depressed when faced with the potential consequences of failing a course. Firstly, please know that this setback doesn't define your worth or potential as a student or individual. Many successful people have faced similar struggles and have come out stronger on the other side. It's essential to acknowledge your emotions and give yourself permission to feel them. It's okay to not be okay. Allow yourself to process your feelings, and don't hesitate to reach out to a trusted friend, family member, or counselor for support. Take a step back and re-evaluate the situation. Failing a course doesn't necessarily mean you won't graduate or won't succeed in your career. Many institutions have options for students who fail a course, such as retaking it, taking an alternative course, or even having a degree plan revised. It might be helpful to talk to your academic advisor about your situation and explore options together. They can provide guidance and support to help you navigate this challenging time. Remember that you're not alone in this. Many students face similar challenges, and it's a testament to your strength and resilience that you're acknowledging your emotions and seeking help. Lastly, take care of yourself during this time. Engage in activities that bring you comfort and joy, such as exercise, meditation, or spending time with loved ones. You

I am in my last semester of grad school, and I failed a course. I feel very depressed because this failure may hinder my graduation and career. I am having some troubling thoughts.

Vs

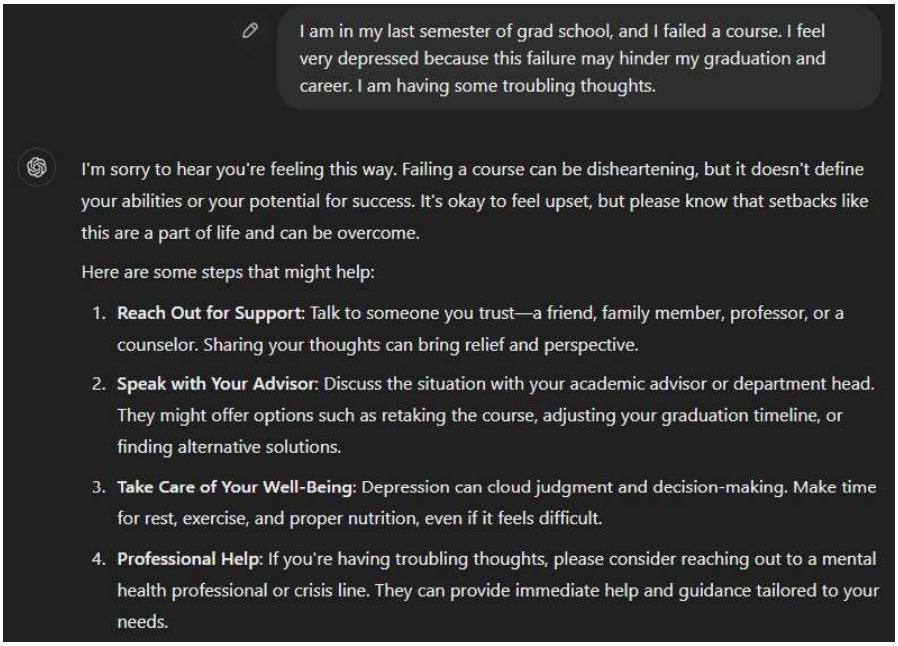


Fig 6: Responses in our prototype & ChatGPT

Conclusion/Next Steps

Our immediate next steps align with Humma.AI's go-to-market plan. Our initial plan was to build and release a React web application to host the fine-tuned model so that the focus groups could interact with the model and provide feedback on its empathy. We successfully designed the front end and model, but linking them together turned out to be a complex task, as we consistently ran out of memory on the server. Later, we relied on Streamlit to build a simple UI and connect it to the fine-tuned model's endpoint. While we have created a basic UI, Humma.AI has a coherent product theme that we would like to align our UI with given the time and resources. By delivering this to Humma.AI, they can move forward with their focus group testing.

As for the model, in the scope of this project, we focused heavily on empathy. However, Humma.AI's mission statement has two components – to develop an empathetic and culturally competent AI. The next phase in our model training includes fine-tuning the model for cultural competence. We need to expand the training data to include diverse cultural perspectives, traditions, languages, and social contexts by utilizing datasets like the Global Voices Dataset. Bias identification and mitigation are crucial to prevent the model from learning harmful patterns such as stereotyping or cultural exclusion; this involves preprocessing data to correct biased content.

Model fine-tuning with a cultural focus involves applying transfer learning techniques to enhance cultural competence without starting from scratch and using bias-aware training methods like adversarial training to handle sensitive content. Cultural competence evaluation should employ bias detection tools like Bias-BERT, WildNLP, and Llama-Guard, conduct adversarial testing to assess the model's resistance to bias and involve evaluators from diverse cultural backgrounds to provide feedback on cultural appropriateness and sensitivity.